5.6 Antiderivatives; Differential Equations

Name _____ Period ____ Date _____

AP Calculus 5.6 (30-minute plan) · Student sheet

I can…

- find all the antiderivatives of a function.
- solve a differential equation with an initial condition.
- find velocity and position from acceleration.

Vocabulary · fill in the missing word

Antiderivative — a _____ F with $F'(x) = f(x)$

General antiderivative — $F(x) + C$, all the _____ of f

Differential equation — an equation that contains a _____, like $\frac{dy}{dx} = f(x)$

Initial condition — a _____ value like $y(0) = 4$, used to find C

Part 1 · Notes and examples (fill in each note, then work the example)

All antiderivatives: if $F' = f$, then every antiderivative of f is $F(x) +$ _____.

Example 1

Find all the antiderivatives of $f(x) = \cos x$.

Power Rule for antiderivatives: $x^n \rightarrow$ _____ + C, for $n \neq -1$.

Example 2

Find all the antiderivatives of $f(x) = 6x^2 - 4x + 5$.

Initial condition: substitute the known value to find _____.

Motion: v is an antiderivative of a, and s is an antiderivative of _____.

Example 3

Solve $\frac{dy}{dx} = 2x + 3$ with $y(0) = 4$.

Part 2 · Partner practice

Partner practice

(AP® Practice) Find the solution of $f'(x) = \frac{3}{x}$, $x > 0$, with $f(e) = 7$.

(A) $f(x) = 3 \ln x + 4$ (B) $f(x) = 3 \ln x - 4$

(C) $f(x) = \ln x + 6$ (D) $f(x) = 3 \ln x + 7$